Lecture Notes on Optimization and Optimal Control

Module 5: Vector Space Convex Optimization

Quanyan Zhu

February 3, 2026

This lecture presents a unified treatment of convex optimization in abstract vector spaces, with duality and saddle-point theory as the central organizing principles. Linear functionals, dual spaces, and cones are introduced to formalize generalized inequality constraints and to extend classical finite-dimensional results to infinite-dimensional settings. Within this cone-based framework, general convex programs admit saddle-point characterizations and strong duality under Slater-type regularity, without reliance on differentiability or constraint qualifications.

When the problem data are Fréchet differentiable, these global results refine naturally into first-order optimality conditions that generalize the classical Karush–Kuhn–Tucker system and clarify its origin in convex duality rather than algebraic rank assumptions. Finite-dimensional optimization emerges as a special case of a broader theory applicable to semidefinite programming, integral constraints, optimal control, and variational problems in function spaces.

1 Linear Functionals and Dual Spaces

This section reviews basic concepts from functional analysis that are essential for convex duality theory. In particular, we introduce linear functionals, continuity and boundedness, and dual spaces, emphasizing concrete examples that illustrate how abstract definitions arise naturally in optimization and control.

Linear functionals. Let Z be a real vector space. A functional $f : Z \to \mathbb{R}$ is called *linear* if for all $z_1, z_2 \in Z$ and all scalars $\alpha, \beta \in \mathbb{R}$,

$$f(\alpha z_1 + \beta z_2) = \alpha f(z_1) + \beta f(z_2).$$

Linear functionals represent measurements, costs, or constraints that depend linearly on the decision variable.

Example 1.1 (Finite-dimensional linear functionals)**.** Let $Z = \mathbb{R}^n$ and $z = (z_1, \ldots, z_n)$. Any functional of the form

$$f(z) = \sum_{i=1}^{n} \eta_i z_i, \qquad \eta \in \mathbb{R}^n,$$

is linear. Conversely, every linear functional on $\mathbb{R}^n$ admits such a representation. Thus, linear functionals on $\mathbb{R}^n$ are in one-to-one correspondence with vectors in $\mathbb{R}^n$.

Example 1.2 (Evaluation functional)**.** Let $Z = C[0,1]$, the space of continuous functions on $[0,1]$. The functional

$$f(x) = x\left(\tfrac{1}{2}\right)$$

is linear. This example shows that linear functionals need not involve integrals or sums; point evaluation is a linear operation.

Example 1.3 (Integral functional)**.** Let $Z = L^2[0,1]$. For a fixed function $y \in L^2[0,1]$, define

$$f(x) = \int_0^1 x(t)y(t)\,dt.$$

This functional is linear in x and arises naturally in optimal control and variational problems.

Continuity and boundedness. When Z is equipped with a norm, not every linear functional is suitable for analysis or optimization. We therefore focus on linear functionals that are continuous.

A linear functional f on a normed space Z is said to be *bounded* if there exists a constant $M \geq 0$ such that

$$|f(z)| \leq M\|z\| \quad \text{for all } z \in Z.$$

The smallest such constant is called the *operator norm* of f and is denoted

$$\|f\| := \sup_{\|z\| \leq 1} |f(z)|.$$

A fundamental and useful fact is that for linear functionals, boundedness and continuity are equivalent: a linear functional is continuous if and only if it is bounded.

Example 1.4 (Boundedness via Cauchy–Schwarz)**.** Let $Z = L^2[0,1]$ and $f(x) = \int_0^1 x(t)y(t)\,dt$ with $y \in L^2[0,1]$. By the Cauchy–Schwarz inequality,

$$|f(x)| \leq \|x\|_2 \|y\|_2.$$

Hence f is bounded and $\|f\| = \|y\|_2$.

Dual spaces. The set of all bounded linear functionals on a normed space Z forms a vector space, called the *dual space* of Z and denoted by Z^*. Equipped with the operator norm

$$\|z^*\| := \sup_{\|z\| \leq 1} |z^*(z)|,$$

the dual space Z^* is itself a normed space. Moreover, Z^* is always complete, even when Z is not; hence Z^* is a Banach space.

Dual spaces provide the natural setting for Lagrange multipliers, adjoint variables, and sensitivity information in optimization. In particular, first- order optimality conditions are expressed through elements of the dual space.

Example 1.5 (Dual of $\mathbb{R}^n$)**.** The dual space $(\mathbb{R}^n)^*$ can be identified with $\mathbb{R}^n$ itself. Under this identification, every linear functional $z^* \in (\mathbb{R}^n)^*$ acts on $z \in \mathbb{R}^n$ via the Euclidean inner product

$$\langle z, z^* \rangle = z^\top z^*.$$

Thus, in finite-dimensional spaces, dual variables can be represented as vectors, and no distinction is made between primal and dual spaces at the level of sets.

Example 1.6 (Gradients as dual elements in finite dimensions)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at x. The directional derivative of f at x in the direction d is

$$\delta f(x; d) = \nabla f(x)^\top d.$$

For fixed x, the mapping $d \mapsto \delta f(x; d)$ is linear and continuous in d. Hence,

$$\nabla f(x) \in (\mathbb{R}^n)^*.$$

This observation explains why gradients appear naturally as multipliers in first-order optimality conditions.

Example 1.7 (Gradients in Hilbert spaces)**.** Let Z be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|z\| = \langle z, z \rangle^{1/2}$. Let $f : Z \to \mathbb{R}$ be Fréchet differentiable at a point $x \in Z$.

By definition, Fréchet differentiability means that there exists a bounded linear functional $Df(x) \in Z^*$ such that

$$\lim_{\|d\| \to 0} \frac{f(x+d) - f(x) - Df(x)[d]}{\|d\|} = 0.$$

Since $Df(x)$ is a bounded linear functional on the Hilbert space Z, the Riesz representation theorem guarantees the existence of a unique element $\nabla f(x) \in Z$ satisfying

$$Df(x)[d] = \langle \nabla f(x), d \rangle, \qquad \forall d \in Z.$$

Equivalently, the directional derivative of f at x in the direction d can be written as

$$\delta f(x; d) = \langle \nabla f(x), d \rangle, \qquad \forall d \in Z.$$

Thus, in a Hilbert space, the gradient $\nabla f(x)$ provides a concrete representation of the derivative as an element of the primal space Z, while simultaneously representing an element of the dual space Z^* via the inner product pairing. This identification is canonical and does not depend on a choice of coordinates. The gradient $\nabla f(x)$ is the unique vector that encodes all first-order variation information of f at x. In particular, the condition

$$\nabla f(x) = 0$$

is equivalent to the vanishing of the derivative $Df(x)$ and therefore characterizes unconstrained minimizers of convex functionals on Hilbert spaces.

Consider the quadratic functional

$$f(z) = \tfrac{1}{2}\|z\|^2, \qquad z \in Z.$$

A direct computation yields

$$\delta f(z; d) = \langle z, d \rangle,$$

so that $\nabla f(z) = z$. In this case, the gradient coincides with the Riesz representative of the linear

functional $d \mapsto \langle z, d \rangle$, making the duality between Z and Z^* completely explicit.

This identification of gradients with dual elements is fundamental in infinite-dimensional optimization, optimal control, and variational problems, where adjoint variables and Lagrange multipliers naturally live in dual spaces.

Example 1.8 (Integral functionals and duality)**.** Let $Z = L^2[a, b]$ and define

$$f(x) = \frac{1}{2} \int_a^b x(t)^2 \, dt.$$

The Fréchet derivative at x is

$$\delta f(x; d) = \int_a^b x(t) d(t) \, dt.$$

Thus, the gradient of f at x is the function x itself, viewed as an element of the dual space $(L^2[a, b])^* \cong L^2[a, b]$. This identification is central in optimal control and calculus of variations.

ℓ^p and L^p spaces. For $1 \leq p < \infty$, the space ℓ^p consists of sequences $x = \{x_i\}_{i=1}^{\infty}$ satisfying

$$\sum_{i=1}^{\infty} |x_i|^p < \infty,$$

with norm $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$. The space ℓ^∞ consists of bounded sequences with the supremum norm.

Similarly, for an interval $[a, b]$, the space $L^p[a, b]$ consists of measurable functions satisfying

$$\int_a^b |x(t)|^p \, dt < \infty,$$

with $L^\infty[a, b]$ defined via the essential supremum.

Hölder's inequality and duality. Let $1 \leq p < \infty$ and define the conjugate exponent q by $1/p + 1/q = 1$. Hölder's inequality states that for $x \in \ell^p$ and $y \in \ell^q$,

$$\sum_{i=1}^{\infty} |x_i y_i| \leq \|x\|_p \|y\|_q,$$

and similarly for $x \in L^p[a, b]$ and $y \in L^q[a, b]$,

$$\int_a^b |x(t) y(t)| \, dt \leq \|x\|_p \|y\|_q.$$

Example 1.9 (Dual of ℓ^p)**.** For $1 < p < \infty$, every bounded linear functional on ℓ^p admits the representation

$$f(x) = \sum_{i=1}^{\infty} x_i y_i$$

for a unique $y \in \ell^q$, and $\|f\| = \|y\|_q$. Thus,

$$(\ell^p)^* \cong \ell^q.$$

Example 1.10 (Dual of L^p)**.** For $1 < p < \infty$, every bounded linear functional on $L^p[a,b]$ can be written as

$$f(x) = \int_a^b x(t)y(t)\,dt$$

for a unique $y \in L^q[a,b]$. This representation is fundamental in infinite- dimensional optimization, where multipliers are functions rather than vectors.

2 Cones

By introducing a cone that defines positivity in a vector space, it is possible to formulate inequality relations and convexity concepts in abstract vector spaces. This provides the foundation for vector optimization, variational inequalities, and duality theory.

2.1 Positive and Negative Cones

Let X be a real vector space.

Definition 2.1 (Positive cone and induced order)**.** Let $P \subset X$ be a convex cone, i.e.,

$$x, y \in P,\ \alpha, \beta \geq 0 \quad \Longrightarrow \quad \alpha x + \beta y \in P.$$

For $x, y \in X$, we define

$$x \succeq y \quad \Longleftrightarrow \quad x - y \in P.$$

The cone P is called the *positive cone* in X. The cone

$$N := -P$$

is called the *negative cone*, and we write $y \preceq x$ if $y - x \in N$.

Example 2.2 (Positive orthant in $\mathbb{R}^n$)**.** In $X = \mathbb{R}^n$, the cone

$$P := \{x = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n : \xi_i \geq 0,\ i = 1, \ldots, n\}$$

defines the standard componentwise partial order on $\mathbb{R}^n$.

Example 2.3 (Function spaces)**.** Let X be a space of real-valued functions on an interval $[t_1, t_2]$. A natural positive cone is

$$P := \{f \in X : f(t) \geq 0 \text{ for all } t \in [t_1, t_2]\}.$$

The induced relation $\succeq$ is reflexive and transitive. Indeed, $x \succeq x$ since $0 \in P$, and if $x \succeq y$ and $y \succeq z$ then $x - z = (x - y) + (y - z) \in P$.

2.2 Closed Cones and Interior Points

In normed spaces, it is often important that the positive cone be closed.

Definition 2.4 (Strict positivity)**.** Let X be a normed space and $P \subset X$ a positive cone. We write

$$x \succ 0$$

if x belongs to the interior of P.

The existence of interior points is crucial for separation arguments.

Example 2.5 (Lack of interior in L^1)**.** Let $X = L^1[t_1, t_2]$ and let P be the cone of nonnegative functions. Then P has empty interior.

Example 2.6 (Interior points in $C[t_1, t_2]$)**.** Let $X = C[t_1, t_2]$ equipped with the uniform norm, and let P be the cone of nonnegative continuous functions. In this case, P possesses interior points.

2.3 Dual Cones

Let X be a normed space with dual space X^*.

Definition 2.7 (Dual cone)**.** Given a positive cone $P \subset X$, the *dual cone* $P^{\oplus} \subset X^*$ is defined by

$$P^{\oplus} := \{x^* \in X^* : \langle x, x^* \rangle \geq 0 \text{ for all } x \in P\}.$$

Example 2.8 (Finite-dimensional case)**.** If $X = \mathbb{R}^n$ and P is the positive orthant, then $P^{\oplus}$ consists of all linear functionals represented by vectors with nonnegative components.

Example 2.9 (Continuous functions)**.** If $X = C[t_1, t_2]$ and P is the cone of nonnegative continuous functions, then $P^{\oplus}$ consists of all linear functionals represented by functions of bounded variation that are nondecreasing on $[t_1, t_2]$.

The dual cone $P^{\oplus}$ is always closed, even when P is not.

2.4 Characterization via Duality

Proposition 2.10 (Dual characterization of positivity)**.** *Let X be a normed space and let $P \subset X$ be a closed positive cone. If $x \in X$ satisfies*

$$\langle x, x^* \rangle \geq 0 \quad \textit{for all } x^* \in P^{\oplus},$$

then $x \in P$.

Proof. Assume $x \notin P$. Since P is closed and convex, the separating hyperplane theorem implies that there exists $x^* \in X^*$ such that

$$\langle x, x^* \rangle < \langle p, x^* \rangle \quad \text{for all } p \in P.$$

Because P is a cone, $\langle p, x^* \rangle$ cannot be negative; otherwise scaling p would violate the inequality. Hence $x^* \in P^{\oplus}$. Moreover,

$$\inf_{p \in P} \langle p, x^* \rangle = 0,$$

which implies $\langle x, x^* \rangle < 0$, contradicting the assumption. □

2.5 Convex Mappings with Respect to a Cone

The generalized notion of inequality allows convexity to be defined for vector-valued mappings.

Definition 2.11 (Cone-convex mapping)**.** Let X and Z be vector spaces, and let $P \subset Z$ be a specified positive cone. A mapping $G : X \to Z$ is said to be *convex with respect to P* if:

1. its domain $\Omega \subset X$ is convex, and

2. for all $x_1, x_2 \in \Omega$ and all $\alpha \in (0,1)$,

$$G(\alpha x_1 + (1-\alpha)x_2) \preceq \alpha G(x_1) + (1-\alpha)G(x_2).$$

Remark 2.12. Convexity of a mapping is not an intrinsic property of the function alone; it depends on the choice of the positive cone in the range space.

We illustrate the notion of cone-convexity through several representative examples. These examples demonstrate how classical convexity, inequality constraints, matrix inequalities, and integral constraints all arise as special cases of convex mappings with respect to an appropriate cone.

Example 2.13 (Scalar-valued convex functions)**.** Let $Z = \mathbb{R}$ and let $P = \mathbb{R}_+$. A mapping $G : X \to \mathbb{R}$ is convex with respect to P if and only if

$$G(\alpha x_1 + (1-\alpha)x_2) \leq \alpha G(x_1) + (1-\alpha)G(x_2),$$

which is precisely the standard definition of convexity for real-valued functions. Thus, scalar convex functions are exactly cone-convex mappings with respect to the nonnegative real cone.

Example 2.14 (Componentwise convex inequality constraints)**.** Let $X = \mathbb{R}^n$, $Z = \mathbb{R}^m$, and $P = \mathbb{R}^m_+$. Define

$$G(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{pmatrix},$$

where each $g_i : \mathbb{R}^n \to \mathbb{R}$ is a convex function. Then G is convex with respect to P if and only if each component g_i is convex. The constraint $G(x) \preceq 0$ is equivalent to the system of scalar inequalities $g_i(x) \leq 0$, $i = 1, \ldots, m$. This example recovers standard finite-dimensional convex programs with multiple inequality constraints.

Example 2.15 (Matrix-valued convex mappings and LMIs)**.** Let $Z = \mathbb{S}^n$, the space of symmetric $n \times n$ matrices, equipped with the cone of positive semidefinite matrices $\mathbb{S}^n_+$. Define

$$G(x) = A_0 + \sum_{i=1}^{k} x_i A_i, \qquad A_i \in \mathbb{S}^n.$$

Then G is affine and hence convex with respect to $\mathbb{S}^n_+$. The constraint $G(x) \preceq 0$ is a linear matrix inequality (LMI), which forms the basic feasibility condition in semidefinite programming.

Example 2.16 (Nonlinear matrix-valued convex mappings)**.** Let $Z = \mathbb{S}^n$ with cone $\mathbb{S}^n_+$, and define

$$G(x) = \begin{pmatrix} x & 0 \\ 0 & x^2 \end{pmatrix}, \qquad x \in \mathbb{R}.$$

Since the mapping $x \mapsto x^2$ is convex and the matrix order is preserved componentwise, G is convex with respect to $\mathbb{S}^n_+$. This example shows that cone-convexity is not limited to affine mappings.

Example 2.17 (Integral inequality constraints)**.** Let $X = L^2[0, T]$, $Z = \mathbb{R}$, and $P = \mathbb{R}_+$. Define

$$G(u) = \int_0^T g(u(t))\, dt - b,$$

where $g : \mathbb{R} \to \mathbb{R}$ is convex and $b \in \mathbb{R}$ is fixed. Then G is convex with respect to P. This example illustrates how integral constraints in optimal control naturally fit into the cone-convex framework.

Example 2.18 (Vector-valued integral constraints)**.** Let $X = L^2[0, T]$, $Z = \mathbb{R}^m$, and $P = \mathbb{R}^m_+$. Define

$$G(u) = \begin{pmatrix} \int_0^T g_1(u(t))\, dt \\ \vdots \\ \int_0^T g_m(u(t))\, dt \end{pmatrix},$$

where each g_i is convex. Then G is convex with respect to P, and the constraint $G(u) \preceq 0$ represents multiple integral inequalities.

Example 2.19 (Same mapping, different cone, different convexity)**.** Let $G : \mathbb{R} \to \mathbb{R}^2$ be defined by $G(x) = (x, -x^2)$. We show that the convexity of G depends on the choice of the positive cone in the range space.

Case 1: $P = \mathbb{R}^2_+$. Recall that G is convex with respect to P if and only if, for all $x_1, x_2 \in \mathbb{R}$ and all $\alpha \in (0, 1)$,

$$G(\alpha x_1 + (1-\alpha)x_2) \preceq \alpha G(x_1) + (1-\alpha)G(x_2),$$

where $\preceq$ denotes the componentwise order induced by $\mathbb{R}^2_+$.

Evaluating both sides, we obtain

$$G(\alpha x_1 + (1-\alpha)x_2) = \big(\alpha x_1 + (1-\alpha)x_2,\ -(\alpha x_1 + (1-\alpha)x_2)^2\big),$$

and

$$\alpha G(x_1) + (1-\alpha)G(x_2) = \big(\alpha x_1 + (1-\alpha)x_2,\ -\alpha x_1^2 - (1-\alpha)x_2^2\big).$$

The first components are equal. For the second component, the required inequality is

$$-(\alpha x_1 + (1-\alpha)x_2)^2 \leq -\alpha x_1^2 - (1-\alpha)x_2^2,$$

which is equivalent to

$$(\alpha x_1 + (1-\alpha)x_2)^2 \geq \alpha x_1^2 + (1-\alpha)x_2^2.$$

This inequality is violated for any $x_1 \neq x_2$, since the square function is strictly convex. For example, taking $x_1 = -1$, $x_2 = 1$, and $\alpha = \frac{1}{2}$, we obtain

$$(\tfrac{1}{2}(-1) + \tfrac{1}{2}(1))^2 = 0 < 1 = \tfrac{1}{2}(1) + \tfrac{1}{2}(1).$$

Hence the defining inequality for cone-convexity fails, and G is *not* convex with respect to $\mathbb{R}^2_+$.

Case 2: $P = \{(u,v) \in \mathbb{R}^2 \mid u \geq 0\}$.

In this case, the partial order induced by P depends only on the first component. That is,

$$(u_1, v_1) \preceq (u_2, v_2) \quad \Longleftrightarrow \quad u_1 \leq u_2,$$

with no restriction on the second component. For all $x_1, x_2 \in \mathbb{R}$ and $\alpha \in (0,1)$, we again compute

$$G(\alpha x_1 + (1-\alpha)x_2) = \big(\alpha x_1 + (1-\alpha)x_2,\ -(\alpha x_1 + (1-\alpha)x_2)^2\big),$$

and

$$\alpha G(x_1) + (1-\alpha)G(x_2) = \big(\alpha x_1 + (1-\alpha)x_2,\ -\alpha x_1^2 - (1-\alpha)x_2^2\big).$$

Since the first components are identical, we have

$$G(\alpha x_1 + (1-\alpha)x_2) \preceq \alpha G(x_1) + (1-\alpha)G(x_2)$$

for all x_1, x_2 and $\alpha \in (0,1)$, regardless of the behavior of the second component.

Thus G is convex with respect to the cone $P = \{(u,v) \mid u \geq 0\}$.

3 General Convex Programming in Vector Spaces

This section develops a general formulation of convex optimization problems posed on abstract vector spaces. The objective is to provide a unified framework that encompasses both finite-dimensional convex programs and infinite-dimensional optimization problems arising in functional analysis, optimal control, and variational problems. The formulation emphasizes geometric structure through ordered vector spaces and cone-induced inequalities, which serve as the foundation for duality and saddle-point theory.

3.1 Problem Statement

Let X be a real vector space and let $\Omega \subset X$ be a nonempty convex set. We consider optimization problems of the form

$$\begin{aligned} \min_{x \in \Omega} \quad & f(x) \\ \text{subject to} \quad & G(x) \preceq 0, \end{aligned} \tag{3.1}$$

where the objective function $f : \Omega \to \mathbb{R}$ is a convex functional and the constraint mapping $G : \Omega \to Z$ takes values in a normed vector space Z.

In order to interpret the vector inequality constraint in (3.1), the space Z is equipped with a closed, convex cone $P \subset Z$, called the *positive cone*. This cone induces a partial order on Z according to

$$z_1 \succeq z_2 \quad \Longleftrightarrow \quad z_1 - z_2 \in P. \tag{3.2}$$

The corresponding negative cone is $-P$, and the notation $z_1 \preceq z_2$ is used when $z_2 - z_1 \in P$. Through this construction, the inequality $G(x) \preceq 0$ is interpreted as the requirement that $G(x)$ lie in the negative cone $-P$.

The cone-induced order provides a generalized notion of inequality that extends the familiar componentwise ordering in $\mathbb{R}^m$ and allows constraints to be expressed in abstract spaces, including spaces of functions or operators.

3.1.1 Cone-Convex Constraint Mappings

The constraint mapping $G : \Omega \to Z$ is assumed to be convex with respect to the cone P. Specifically, for all $x_1, x_2 \in \Omega$ and all $\alpha \in (0, 1)$, the mapping satisfies

$$G(\alpha x_1 + (1 - \alpha)x_2) \preceq \alpha G(x_1) + (1 - \alpha)G(x_2). \tag{3.3}$$

This condition generalizes the usual notion of convexity for real-valued functions to vector-valued mappings by replacing scalar inequalities with cone-induced partial orders. Convexity of G with respect to P ensures that the feasible set of (3.1) is convex and that the constraint structure is compatible with separation and duality arguments.

Problem (3.1) is referred to as the *general convex programming problem*. It provides a common abstraction for classical inequality-constrained convex programs in finite dimensions and for infinite-dimensional optimization problems posed in Banach or Hilbert spaces.

3.1.2 Dual Cone and Lagrangian

Let Z^* denote the dual space of Z, consisting of all continuous linear functionals on Z. Associated with the positive cone $P \subset Z$ is the dual cone

$$P^{\oplus} := \{z^* \in Z^* \mid \langle z, z^* \rangle \geq 0 \text{ for all } z \in P\}. \tag{3.4}$$

Elements of $P^{\oplus}$ are precisely those linear functionals that preserve the cone-induced order, in the sense that they assign nonnegative values to all elements of the positive cone.

The Lagrangian associated with the general convex program (3.1) is defined as

$$L(x, z^*) := f(x) + \langle G(x), z^* \rangle, \qquad z^* \in P^{\oplus}. \tag{3.5}$$

This construction extends the classical Lagrangian for scalar inequality constraints by replacing nonnegative multipliers with elements of the dual cone. The pairing $\langle G(x), z^* \rangle$ represents the action of a constraint-weighting functional on the vector-valued constraint, thereby encoding feasibility and optimality information in a scalar quantity.

3.1.3 Slater Condition

The existence of meaningful Lagrange multipliers requires a regularity assumption ensuring that the feasible set is not degenerate relative to the cone-induced order.

Definition 3.1 (Slater Condition)**.** Problem (3.1) is said to satisfy Slater's condition if there exists a point $\bar{x} \in \Omega$ such that

$$G(\bar{x}) \prec 0, \tag{3.6}$$

that is, $G(\bar{x})$ lies in the interior of the negative cone $-P$.

This condition asserts the existence of a strictly feasible point and plays a central role in convex duality theory. Geometrically, it ensures that the constraint set admits interior points relative to the cone order, which in turn guarantees the existence of nontrivial supporting hyperplanes and enables the derivation of saddle-point characterizations and strong duality results.

3.1.4 Illustrative Examples

We illustrate the general convex programming formulation through several representative examples. These examples are chosen to demonstrate how the abstract cone-based framework specializes to familiar finite-dimensional problems, extends naturally to infinite-dimensional settings, and accommodates matrix-valued inequality constraints. In each case, the structure of the dual cone clarifies the nature of the associated Lagrange multipliers.

Example 3.2 (Finite-dimensional inequality-constrained convex program)**.** Let $X = \mathbb{R}^n$ and let $\Omega \subset \mathbb{R}^n$ be a convex set. Let the objective function $f : \Omega \to \mathbb{R}$ be convex, and define the constraint mapping $G : \Omega \to \mathbb{R}^m$ componentwise by

$$G(x) = (g_1(x), \ldots, g_m(x)),$$

where each component function $g_i : \Omega \to \mathbb{R}$ is convex.

Equip $Z = \mathbb{R}^m$ with the positive orthant

$$P := \mathbb{R}^m_+ = \{z \in \mathbb{R}^m : z_i \geq 0,\ i = 1, \ldots, m\}.$$

The cone P induces the standard componentwise partial order on $\mathbb{R}^m$. Under this order, the vector inequality

$$G(x) \preceq 0$$

is equivalent to the system of scalar constraints

$$g_i(x) \leq 0, \quad i = 1, \ldots, m.$$

Convexity of each g_i implies that the mapping G is convex with respect to the cone P in the sense of (3.3). As a result, the general convex programming problem (3.1) reduces exactly to the classical finite-dimensional convex optimization problem with inequality constraints.

In this setting, the dual space satisfies $Z^* \cong \mathbb{R}^m$ under the standard inner product, and the dual cone $P^\oplus$ coincides with $\mathbb{R}^m_+$. The requirement that dual variables belong to $P^\oplus$ therefore recovers the familiar nonnegativity condition on Lagrange multipliers. The Lagrangian (3.5) takes the form

$$L(x, \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x), \qquad \lambda_i \geq 0,$$

demonstrating that the classical theory of Lagrange multipliers arises as a special case of the cone-based formulation.

Example 3.3 (Integral inequality constraint in a function space)**.** Let $X = L^2([0, T])$ and let $\Omega \subset L^2([0, T])$ be a convex set of admissible functions. Consider the objective functional

$$f(x) := \int_0^T \ell(x(t))\, dt,$$

where $\ell : \mathbb{R} \to \mathbb{R}$ is convex. Define the constraint mapping $G : \Omega \to \mathbb{R}$ by

$$G(x) := \int_0^T x(t)\, dt - b,$$

where $b \in \mathbb{R}$ is a prescribed constant.

Let $Z = \mathbb{R}$ equipped with the positive cone $P = \mathbb{R}_+$. The constraint $G(x) \preceq 0$ then corresponds to the integral inequality

$$\int_0^T x(t)\, dt \leq b.$$

Since G is affine, it is convex with respect to P, and the resulting problem fits the general convex programming formulation (3.1).

Slater's condition holds provided there exists a function $\bar{x} \in \Omega$ such that

$$\int_0^T \bar{x}(t)\, dt < b.$$

In this case, strong duality applies. The dual space Z^* is again identified with $\mathbb{R}$, and the dual cone $P^\oplus$ is $\mathbb{R}_+$. The associated dual variable is therefore a scalar multiplier $\lambda \geq 0$, and the Lagrangian becomes

$$L(x, \lambda) = \int_0^T \ell(x(t))\, dt + \lambda \left(\int_0^T x(t)\, dt - b \right).$$

This example illustrates how infinite-dimensional optimization problems with integral con-

straints admit Lagrange multipliers of the same algebraic form as in finite dimensions, while still fitting within the abstract cone-based theory.

Example 3.4 (Matrix inequality constraints and semidefinite programming)**.** Let $X = \mathbb{R}^n$ and let $\Omega \subset \mathbb{R}^n$ be convex. Suppose that the constraint mapping

$$G : \Omega \to \mathbb{S}^m$$

takes values in the space $\mathbb{S}^m$ of real symmetric $m \times m$ matrices. Equip $Z = \mathbb{S}^m$ with the positive semidefinite cone

$$P := \mathbb{S}^m_+ = \{M \in \mathbb{S}^m : M \succeq 0\}.$$

The constraint

$$G(x) \preceq 0$$

is then interpreted in the Löwner order, meaning that $G(x)$ is negative semidefinite. If G is convex with respect to the cone $\mathbb{S}^m_+$, then (3.1) defines a semidefinite program.

The dual space of $\mathbb{S}^m$ is naturally identified with $\mathbb{S}^m$ itself via the trace pairing

$$\langle M, Y \rangle := \operatorname{trace}(MY).$$

Under this identification, the dual cone $P^\oplus$ coincides with $\mathbb{S}^m_+$. Thus, the dual variables are themselves positive semidefinite matrices, and the Lagrangian takes the form

$$L(x, Y) = f(x) + \operatorname{trace}(G(x)Y), \qquad Y \succeq 0.$$

Slater's condition requires the existence of a strictly feasible point $\bar{x} \in \Omega$ such that $G(\bar{x})$ is negative definite. When this condition is satisfied, the problem admits a saddle-point characterization and strong duality holds. This example highlights how matrix inequality constraints and semidefinite programs arise naturally within the general convex programming framework.

3.2 Saddle-Point Characterization and Strong Duality

Theorem 3.5 (Saddle-Point and Strong Duality)**.** *Assume that f and G are convex and that Slater's condition* (3.6) *holds. A point $x^\star \in \Omega$ is an optimal solution of* (3.1) *if and only if there exists $z^{\star\star} \in P^\oplus$ such that*

$$L(x^\star, z^*) \le L(x^\star, z^{\star\star}) \le L(x, z^{\star\star}), \quad \forall x \in \Omega,\ \forall z^* \in P^\oplus. \tag{3.7}$$

Moreover, strong duality holds:

$$\inf_{x\in\Omega}\sup_{z^*\in P^{\oplus}} L(x,z^*) = \sup_{z^*\in P^{\oplus}}\inf_{x\in\Omega} L(x,z^*). \tag{3.8}$$

Proof. We prove the equivalence in two directions and then derive strong duality as a consequence. Throughout the proof, recall that the Lagrangian is defined by

$$L(x,z^*) = f(x) + \langle G(x), z^*\rangle, \qquad z^* \in P^{\oplus}.$$

(⇐) Existence of a saddle point implies primal optimality. Assume that there exists a pair $(x^\star, z^{**}) \in \Omega \times P^{\oplus}$ satisfying the saddle-point inequalities (3.7), namely,

$$L(x^\star, z^*) \le L(x^\star, z^{**}) \le L(x, z^{**}), \quad \forall x \in \Omega,\ \forall z^* \in P^{\oplus}.$$

We first show that $x^\star$ must be feasible. Taking $z^* = 0$ in the left inequality gives

$$f(x^\star) = L(x^\star, 0) \le L(x^\star, z^{**}) = f(x^\star) + \langle G(x^\star), z^{**}\rangle,$$

which implies

$$\langle G(x^\star), z^{**}\rangle \ge 0. \tag{3.9}$$

Since $z^{**} \in P^{\oplus}$, we have $\langle z, z^{**}\rangle \ge 0$ for all $z \in P$. If $G(x^\star) \not\preceq 0$, then $G(x^\star)$ would contain a positive component in the cone order, implying $\langle G(x^\star), z^{**}\rangle > 0$ for some $z^{**} \in P^{\oplus}$, which would contradict the saddle-point inequality for large scalar multiples of z^{**}. Hence $G(x^\star) \preceq 0$, and $x^\star$ is feasible.

Next, since $G(x^\star) \preceq 0$ and $z^{**} \in P^{\oplus}$, we also have

$$\langle G(x^\star), z^{**}\rangle \le 0. \tag{3.10}$$

Combining (3.9) and (3.10) yields

$$\langle G(x^\star), z^{**}\rangle = 0.$$

Now let $x \in \Omega$ be any feasible point, i.e. $G(x) \preceq 0$. From the right inequality in (3.7),

$$L(x^\star, z^{**}) \le L(x, z^{**}) = f(x) + \langle G(x), z^{**}\rangle.$$

Since $G(x) \preceq 0$ and $z^{\star\star} \in P^{\oplus}$, the inner product on the right-hand side is nonpositive. Using $L(x^\star, z^{\star\star}) = f(x^\star)$, we conclude

$$f(x^\star) \leq f(x).$$

Because x was arbitrary among feasible points, $x^\star$ is a primal optimal solution of (3.1).

(⇒) Primal optimality implies existence of a saddle point. Assume now that $x^\star \in \Omega$ is a primal optimal solution of (3.1). Define the optimal value

$$\mu^\star := f(x^\star).$$

The key idea is to encode the primal problem geometrically in a higher-dimensional space so that convex separation can be applied. Consider the product space $\mathbb{R} \times Z$ and define the sets

$$\mathcal{A} := \{(r, z) \in \mathbb{R} \times Z \mid \exists x \in \Omega \text{ such that } r \geq f(x),\ z \succeq G(x)\},$$

and

$$\mathcal{B} := \{(r, z) \in \mathbb{R} \times Z \mid r \leq \mu^\star,\ z \preceq 0\}.$$

The set $\mathcal{A}$ is convex because it is the image of the convex set Ω under the convex mapping $x \mapsto (f(x), G(x))$ followed by upward closure with respect to the cone order. The set $\mathcal{B}$ is convex because it is defined by linear inequalities.

By definition of $\mu^\star$, there exists no point that simultaneously satisfies $r \geq f(x)$ with $r \leq \mu^\star$ and $z \succeq G(x)$ with $z \preceq 0$. Hence,

$$\mathcal{A} \cap \mathcal{B} = \emptyset.$$

Slater's condition ensures the existence of a strictly feasible point $\bar{x} \in \Omega$ such that $G(\bar{x}) \prec 0$. This implies that $\mathcal{B}$ contains interior points in $\mathbb{R} \times Z$, which prevents degenerate (vertical) separating hyperplanes and guarantees the applicability of the separating hyperplane theorem.

By the separating hyperplane theorem, there exists a nonzero continuous linear functional $(\alpha, z^{\star\star}) \in \mathbb{R} \times Z^*$ such that

$$\alpha r + \langle z, z^{\star\star} \rangle \geq \alpha r' + \langle z', z^{\star\star} \rangle \tag{3.11}$$

for all $(r, z) \in \mathcal{A}$ and $(r', z') \in \mathcal{B}$.

We now analyze the structure of the separating functional. From the definition of $\mathcal{B}$, inequality (3.11) implies $\alpha \geq 0$ and $\langle z, z^{\star\star} \rangle \geq 0$ for all $z \in P$. Hence $z^{\star\star} \in P^{\oplus}$. Moreover, $\alpha \neq 0$;

otherwise strict feasibility would be violated. We therefore normalize $\alpha = 1$.

With this normalization, inequality (3.11) implies

$$f(x) + \langle G(x), z^{**} \rangle \geq \mu^\star, \quad \forall x \in \Omega,$$

or equivalently,

$$L(x, z^{**}) \geq f(x^\star), \quad \forall x \in \Omega. \tag{3.12}$$

Finally, for any $z^* \in P^\oplus$, feasibility of $x^\star$ implies

$$L(x^\star, z^*) = f(x^\star) + \langle G(x^\star), z^* \rangle \leq f(x^\star) = L(x^\star, z^{**}),$$

while (3.12) gives

$$L(x^\star, z^{**}) \leq L(x, z^{**}), \quad \forall x \in \Omega.$$

Thus $(x^\star, z^{**})$ satisfies the saddle-point inequalities (3.7).

Strong duality. The equivalence established above shows that primal optimal solutions exist if and only if saddle points exist. For any saddle point $(x^\star, z^{**})$,

$$\sup_{z^* \in P^\oplus} \inf_{x \in \Omega} L(x, z^*) = L(x^\star, z^{**}) = \inf_{x \in \Omega} \sup_{z^* \in P^\oplus} L(x, z^*),$$

which establishes the minimax equality (3.8). □

Corollary 3.6 (Complementary Slackness)**.** *Suppose that the assumptions of Theorem 3.5 hold, and let $(x^\star, z^{**}) \in \Omega \times P^\oplus$ be a saddle point of the Lagrangian*

$$L(x, z^*) = f(x) + \langle G(x), z^* \rangle.$$

Then the primal and dual variables satisfy the complementary slackness condition

$$\langle G(x^\star), z^{**} \rangle = 0. \tag{3.13}$$

Proof. Since $(x^\star, z^{**})$ is a saddle point, $x^\star$ is a feasible point of the primal problem, and $z^{**} \in P^\oplus$ is dual feasible. Hence $G(x^\star) \preceq 0$ and, by definition of the dual cone,

$$\langle G(x^\star), z^{**} \rangle \leq 0. \tag{3.14}$$

On the other hand, the left-hand inequality in the saddle-point condition (3.7) holds for all $z^* \in P^{\oplus}$. In particular, evaluating it at $z^* = 0$ yields

$$L(x^\star, 0) \leq L(x^\star, z^{**\star}),$$

or equivalently,

$$f(x^\star) \leq f(x^\star) + \langle G(x^\star), z^{**\star} \rangle. \tag{3.15}$$

This implies

$$\langle G(x^\star), z^{**\star} \rangle \geq 0.$$

Combining (3.14) and (3.15) establishes (3.18). □

3.3 Connection to Sion's Minimax Theorem and Zero-Sum Games

The saddle-point characterization established in Theorem 3.5 can be interpreted as a special case of a general minimax principle. In particular, it is closely related to Sion's minimax theorem and admits a natural interpretation as a zero-sum game between a primal and a dual player.

Sion's minimax theorem applies to a function

$$\Phi : X \times Y \to \mathbb{R}$$

defined on convex sets X and Y, provided that $\Phi(\cdot, y)$ is convex and lower semicontinuous for each fixed y, $\Phi(x, \cdot)$ is concave and upper semicontinuous for each fixed x, and at least one of the sets is compact. Under these conditions, Sion's theorem asserts that

$$\inf_{x \in X} \sup_{y \in Y} \Phi(x, y) = \sup_{y \in Y} \inf_{x \in X} \Phi(x, y).$$

In the present setting, we consider the function

$$\Phi(x, z^*) := L(x, z^*) = f(x) + \langle G(x), z^* \rangle,$$

defined on the product set $\Omega \times P^{\oplus}$. For each fixed $z^* \in P^{\oplus}$, the function $x \mapsto L(x, z^*)$ is convex, since it is the sum of a convex functional and a linear functional. For each fixed $x \in \Omega$, the function $z^* \mapsto L(x, z^*)$ is affine and hence concave over the convex set $P^{\oplus}$.

Thus, the Lagrangian satisfies the structural convex–concave assumptions of Sion's theo-

rem. However, in general, neither Ω nor $P^{\oplus}$ is compact, particularly in infinite-dimensional problems. The role of compactness in Sion's theorem is therefore replaced here by Slater's condition and the separating hyperplane argument used in the proof of Theorem 3.5. In this sense, the saddle-point theorem proved above can be viewed as a noncompact, cone-constrained extension of Sion's minimax theorem tailored to convex optimization problems.

Zero-sum game interpretation. The Lagrangian naturally defines a two-player zero-sum game. The primal player chooses a decision variable $x \in \Omega$ in order to minimize the payoff $L(x, z^*)$, while the dual player chooses a multiplier $z^* \in P^{\oplus}$ in order to maximize the same quantity. The payoff function of the game is therefore given by

$$(x, z^*) \mapsto L(x, z^*).$$

In this interpretation, the primal player attempts to minimize the objective while satisfying the constraints, whereas the dual player penalizes constraint violations through the choice of an order-preserving linear functional. The restriction $z^* \in P^{\oplus}$ ensures that the dual player can only penalize violations in directions consistent with the cone-induced order.

A saddle point $(x^\star, z^{**})$ of the Lagrangian corresponds exactly to a Nash equilibrium of this zero-sum game. The inequalities

$$L(x^\star, z^*) \leq L(x^\star, z^{**}) \leq L(x, z^{**})$$

state that neither player can improve their outcome by unilaterally deviating from $(x^\star, z^{**})$. In particular, the primal player cannot achieve a lower payoff by choosing a different feasible decision, and the dual player cannot increase the payoff by choosing a different multiplier.

The minimax equality (3.8) expresses the equivalence between the value of the game when the primal player moves first and the value when the dual player moves first. Strong duality thus corresponds to the existence of a game value and a pure-strategy equilibrium in this zero-sum game.

3.4 Primal Problem

The saddle-point characterization established in Theorem 3.5 immediately yields a primal-side representation of the optimal value in terms of the Lagrangian. This result is often referred to as the Lagrange multiplier theorem for convex programs.

Corollary 3.7 (Lagrange Multiplier Theorem (Primal Form)). *Let all assumptions of Theorem 3.5 hold, and define the optimal value*

$$\mu_0 := \inf\{f(x) : x \in \Omega,\ G(x) \preceq 0\}. \tag{3.16}$$

Then there exists an element $z_0^* \in P^{\oplus}$ such that

$$\mu_0 = \inf_{x\in\Omega} \{f(x) + \langle G(x), z_0^*\rangle\}. \tag{3.17}$$

Moreover, if the infimum in (3.16) *is attained at some $x_0 \in \Omega$ with $G(x_0) \preceq 0$, then x_0 also attains the infimum in* (3.17) *and satisfies the complementary slackness condition*

$$\langle G(x_0), z_0^*\rangle = 0. \tag{3.18}$$

Proof. By Theorem 3.5, there exists a saddle point $(x^\star, z_0^*) \in \Omega \times P^{\oplus}$ of the Lagrangian such that

$$L(x^\star, z^*) \le L(x^\star, z_0^*) \le L(x, z_0^*) \quad \forall x \in \Omega,\ \forall z^* \in P^{\oplus}.$$

Restricting this inequality to the primal variable yields

$$L(x, z_0^*) \ge L(x^\star, z_0^*) = f(x^\star) = \mu_0, \quad \forall x \in \Omega.$$

Taking the infimum over $x \in \Omega$ gives

$$\inf_{x\in\Omega}\{f(x) + \langle G(x), z_0^*\rangle\} = \mu_0,$$

which proves (3.17).

If x_0 attains the infimum in (3.16), then feasibility implies

$$f(x_0) + \langle G(x_0), z_0^*\rangle \le f(x_0) = \mu_0.$$

Combined with (3.17), this yields

$$\langle G(x_0), z_0^*\rangle = 0,$$

establishing complementary slackness. □

3.5 Dual Problem

The saddle-point characterization also admits a dual-side interpretation. By projecting the saddle-point inequality onto the dual variable, one obtains an explicit formulation of the dual optimization problem and a characterization of its optimal solutions.

Corollary 3.8 (Dual Problem and Dual Optimality)**.** *Let all assumptions of Theorem 3.5 hold, and define the dual function*

$$\psi(z^*) := \inf_{x\in\Omega} \{f(x) + \langle G(x), z^*\rangle\}, \qquad z^* \in P^{\oplus}. \tag{3.19}$$

Then the dual problem is

$$\sup_{z^*\in P^{\oplus}} \psi(z^*). \tag{3.20}$$

Moreover, there exists $z^{\star\star} \in P^{\oplus}$ such that

$$\mu_0 = \sup_{z^*\in P^{\oplus}} \psi(z^*) = \psi(z^{\star\star}), \tag{3.21}$$

and any such $z^{\star\star}$ is a dual optimal solution.

Proof. By Theorem 3.5, there exists a saddle point $(x^\star, z^{\star\star}) \in \Omega \times P^{\oplus}$ satisfying

$$L(x^\star, z^*) \le L(x^\star, z^{\star\star}) \le L(x, z^{\star\star}), \quad \forall x \in \Omega,\ \forall z^* \in P^{\oplus}.$$

Restricting this inequality to the dual variable yields

$$L(x^\star, z^*) \le L(x^\star, z^{\star\star}), \quad \forall z^* \in P^{\oplus}.$$

Since

$$\psi(z^*) = \inf_{x\in\Omega} L(x, z^*) \le L(x^\star, z^*),$$

we obtain

$$\psi(z^*) \le L(x^\star, z^{\star\star}) = f(x^\star) = \mu_0, \quad \forall z^* \in P^{\oplus}.$$

Thus μ_0 is an upper bound on the dual objective.

On the other hand, evaluating the dual function at $z^{\star\star}$ gives

$$\psi(z^{\star\star}) = \inf_{x\in\Omega} L(x, z^{\star\star}) = L(x^\star, z^{\star\star}) = \mu_0,$$

where the second equality follows from the primal-side inequality of the saddle point. Hence z^{**} attains the supremum in (3.20), and (3.21) follows. □

We now collect several fundamental properties of the dual function (3.19) and the associated dual problem (3.20). These properties hold independently of optimality and clarify the structure of the dual problem.

Lemma 3.9 (Concavity of the Dual Function)**.** *The dual function $\psi : P^{\oplus} \to \mathbb{R} \cup \{-\infty\}$ defined by* (3.19) *is concave.*

Proof. For each fixed $x \in \Omega$, the mapping

$$z^* \mapsto f(x) + \langle G(x), z^* \rangle$$

is affine in z^*. The dual function ψ is defined as the pointwise infimum over $x \in \Omega$ of these affine functions. Since the infimum of affine functions is concave, ψ is concave on $P^{\oplus}$. □

Lemma 3.10 (Weak Duality)**.** *For every feasible primal point $x \in \Omega$ with $G(x) \preceq 0$ and every dual feasible point $z^* \in P^{\oplus}$, the inequality*

$$\psi(z^*) \leq f(x) \tag{3.22}$$

holds. In particular,

$$\sup_{z^* \in P^{\oplus}} \psi(z^*) \leq \mu_0.$$

Proof. Let $x \in \Omega$ satisfy $G(x) \preceq 0$, and let $z^* \in P^{\oplus}$. By definition of the dual function,

$$\psi(z^*) = \inf_{y \in \Omega} \{f(y) + \langle G(y), z^* \rangle\} \leq f(x) + \langle G(x), z^* \rangle.$$

Since $z^* \in P^{\oplus}$ and $G(x) \preceq 0$, we have $\langle G(x), z^* \rangle \leq 0$, which yields (3.22). Taking the supremum over $z^* \in P^{\oplus}$ proves the second claim. □

Lemma 3.11 (Upper Envelope Interpretation)**.** *The dual problem* (3.20) *seeks the tightest lower bound on the primal objective among all Lagrangian relaxations. That is,*

$$\sup_{z^* \in P^{\oplus}} \psi(z^*) = \sup_{z^* \in P^{\oplus}} \inf_{x \in \Omega} \{f(x) + \langle G(x), z^* \rangle\}.$$

Proof. For each $z^* \in P^{\oplus}$, the function $x \mapsto f(x) + \langle G(x), z^* \rangle$ defines a relaxation of the primal problem in which constraint violations are penalized linearly. The dual function $\psi(z^*)$ is the

optimal value of this relaxed problem. Maximizing ψ over $P^{\oplus}$ therefore selects the strongest such relaxation, i.e., the largest lower bound on the primal optimal value. $\square$

3.6 Dealing with Equality Constraints

Thus far, we have considered only convex inequality constraints of the form $G(x) \preceq 0$. Suppose now that the problem includes a linear equality constraint

$$H(x) = 0, \tag{3.23}$$

where $H : X \to Y$ is a continuous linear mapping into another vector space Y. In finite dimensions, one may think of $H(x) = Ax - b$.

Formally, the equality constraint (3.23) can be written as the pair of convex inequalities

$$H(x) \preceq 0, \qquad -H(x) \preceq 0,$$

and therefore appears to fit within the framework of inequality constraints. However, this reformulation introduces a fundamental difficulty: there can never exist a point x_1 such that both

$$H(x_1) \prec 0 \quad \text{and} \quad -H(x_1) \prec 0$$

hold simultaneously. Consequently, Slater's condition fails automatically for the combined inequality system.

This failure is not merely technical. From a geometric perspective, the feasible set defined by $H(x) = 0$ lies on the boundary of the ordering cone in every direction, and thus does not admit a strictly feasible interior point. As a result, the separating hyperplane argument underlying the saddle-point theorem cannot be applied directly.

To accommodate equality constraints, one must treat them separately from inequality constraints. Consider the composite problem

$$\begin{aligned} \min_{x \in \Omega} \quad & f(x) \\ \text{s.t.} \quad & G(x) \preceq 0, \\ & H(x) = 0, \end{aligned} \tag{3.24}$$

where $G : \Omega \to Z$ is convex with respect to a cone $P \subset Z$ with nonempty interior, and $H : X \to$

Y is a linear mapping.

In this setting, the appropriate Lagrangian is defined by

$$L(x, z^*, \lambda) = f(x) + \langle G(x), z^* \rangle + \langle H(x), \lambda \rangle, \tag{3.25}$$

where $z^* \in P^{\oplus}$ is the multiplier associated with the inequality constraints, and $\lambda \in Y^*$ is an unrestricted multiplier associated with the equality constraint.

Unlike the inequality multipliers, the equality multipliers are not restricted to lie in a cone. This reflects the fact that equality constraints impose two-sided restrictions and therefore correspond to unconstrained dual variables.

Regularity and duality. A suitable regularity condition for the mixed problem (3.24) is the existence of a point $\bar{x} \in \Omega$ such that

$$G(\bar{x}) \prec 0, \qquad H(\bar{x}) = 0.$$

Under this condition, one can again apply a separating hyperplane argument in an augmented product space and obtain a saddle-point characterization and strong duality for the mixed problem. The resulting optimality conditions generalize the earlier saddle-point theorem and lead, under differentiability assumptions, to the Karush–Kuhn–Tucker conditions.

We emphasize that equality constraints must be handled explicitly in this composite manner; they cannot be absorbed into the inequality framework without destroying the regularity structure required for duality theory.

Theorem 3.12 (Saddle-Point and Strong Duality with Equality Constraints)**.** *Let X be a real vector space and $\Omega \subset X$ a nonempty convex set. Let $f : \Omega \to \mathbb{R}$ be a convex functional. Let $G : \Omega \to Z$ be convex with respect to a closed convex cone $P \subset Z$ with nonempty interior, and let $H : X \to Y$ be a continuous linear mapping.*

Consider the mixed constrained problem

$$\begin{aligned} \min_{x \in \Omega} \quad & f(x) \\ s.t. \quad & G(x) \preceq 0, \\ & H(x) = 0. \end{aligned} \tag{3.26}$$

Assume that there exists a point $\bar{x} \in \Omega$ such that

$$G(\bar{x}) \prec 0, \qquad H(\bar{x}) = 0, \tag{3.27}$$

(i.e., Slater's condition holds for the inequality constraints and the equality constraints are feasible).

Define the Lagrangian

$$L(x, z^*, \lambda) = f(x) + \langle G(x), z^* \rangle + \langle H(x), \lambda \rangle, \qquad z^* \in P^{\oplus},\ \lambda \in Y^*. \tag{3.28}$$

Then a point $x^\star \in \Omega$ is an optimal solution of (3.26) *if and only if there exist multipliers $(z^{**\star}, \lambda^\star) \in P^{\oplus} \times Y^*$ such that*

$$L(x^\star, z^*, \lambda) \leq L(x^\star, z^{**\star}, \lambda^\star) \leq L(x, z^{**\star}, \lambda^\star), \quad \forall x \in \Omega,\ \forall z^* \in P^{\oplus},\ \forall \lambda \in Y^*. \tag{3.29}$$

Moreover, strong duality holds:

$$\inf_{x \in \Omega} \sup_{\substack{z^* \in P^{\oplus} \\ \lambda \in Y^*}} L(x, z^*, \lambda) = \sup_{\substack{z^* \in P^{\oplus} \\ \lambda \in Y^*}} \inf_{x \in \Omega} L(x, z^*, \lambda). \tag{3.30}$$

Corollary 3.13 (Lagrange Multiplier Theorem (Primal Form with Equality Constraints))**.** *Let all assumptions of Theorem 3.12 hold, and define the optimal value*

$$\mu_0 := \inf\{f(x) : x \in \Omega,\ G(x) \preceq 0,\ H(x) = 0\}. \tag{3.31}$$

Assume that μ_0 is finite.

Then there exist multipliers

$$z_0^* \in P^{\oplus}, \qquad \lambda_0 \in Y^*$$

such that

$$\mu_0 = \inf_{x \in \Omega} \left\{ f(x) + \langle G(x), z_0^* \rangle + \langle H(x), \lambda_0 \rangle \right\}. \tag{3.32}$$

Moreover, if the infimum in (3.31) *is attained at some $x_0 \in \Omega$ with*

$$G(x_0) \preceq 0, \qquad H(x_0) = 0,$$

then x_0 also attains the infimum in (3.32) *and satisfies the complementary slackness condition*

$$\langle G(x_0), z_0^* \rangle = 0. \tag{3.33}$$

3.7 First-Order Optimality Conditions

3.7.1 First-Order Optimality for Optimization over Cones

We consider the following cone-constrained optimization problem:

$$\begin{aligned} \min_{x \in X} \quad & f(x) \\ \text{s.t.} \quad & x \in P, \end{aligned} \tag{3.34}$$

where X is a real normed vector space, $P \subset X$ is a convex cone, and $f : X \to \mathbb{R}$ is a convex functional. The feasible set P models one-sided or inequality-type constraints and includes, as special cases, nonnegativity constraints, ordering cones, and tangent cones arising in constrained optimization.

When f is Fréchet differentiable, optimality of a feasible point can be characterized purely in terms of first-order directional derivatives. The following lemma provides a necessary and sufficient condition for optimality over a convex cone.

Lemma 3.14 (First-Order Optimality over a Cone)**.** *Let X be a real normed vector space and let $P \subset X$ be a convex cone. Let $f : X \to \mathbb{R}$ be a Fréchet differentiable convex functional. A point $x_0 \in P$ minimizes f over P if and only if*

$$\delta f(x_0; x) \geq 0, \qquad \forall x \in P, \tag{3.35}$$

$$\delta f(x_0; x_0) = 0. \tag{3.36}$$

Proof. Necessity. Assume that $x_0 \in P$ minimizes f over P. Since P is convex, for any $x \in P$ and any $\alpha \geq 0$ sufficiently small, the point

$$x_0 + \alpha(x - x_0) \in P$$

is feasible. Therefore, the function $\alpha \mapsto f(x_0 + \alpha(x - x_0))$ attains a minimum at $\alpha = 0$, which implies

$$\frac{d}{d\alpha} f\big(x_0 + \alpha(x - x_0)\big)\Big|_{\alpha=0} \geq 0.$$

By definition of the Fréchet directional derivative, this yields

$$\delta f(x_0; x - x_0) \geq 0, \qquad \forall x \in P. \tag{3.37}$$

Choosing $x = \frac{1}{2}x_0 \in P$ in (3.37) gives

$$\delta f(x_0; -\tfrac{1}{2}x_0) \geq 0,$$

which implies

$$\delta f(x_0; x_0) \leq 0. \tag{3.38}$$

Similarly, choosing $x = 2x_0 \in P$ yields

$$\delta f(x_0; x_0) \geq 0.$$

Together, these inequalities imply (3.36). Substituting (3.36) back into (3.37) establishes (3.35).

Sufficiency. Assume that (3.35) and (3.36) hold. By convexity of f, for any $x \in P$ and any $0 < \alpha < 1$,

$$f(x_0 + \alpha(x - x_0)) \leq f(x_0) + \alpha\big(f(x) - f(x_0)\big).$$

Rearranging yields

$$f(x) - f(x_0) \geq \frac{1}{\alpha}\Big[f(x_0 + \alpha(x - x_0)) - f(x_0)\Big].$$

Taking the limit as $\alpha \downarrow 0$ and using Fréchet differentiability, we obtain

$$f(x) - f(x_0) \geq \delta f(x_0; x - x_0).$$

Using (3.35) and (3.36), we conclude

$$f(x) \geq f(x_0), \qquad \forall x \in P,$$

and hence x_0 minimizes f over P. □

Lemma 3.14 generalizes the familiar one-dimensional first-order optimality condition for convex functions to optimization over arbitrary convex cones in normed spaces. Condition (3.35) requires that the directional derivative be nonnegative along all feasible directions of the cone, while condition (3.36) enforces stationarity along the radial direction of the cone. Together,

these conditions capture both interior and boundary optimality in a unified form.

This lemma provides the differential underpinning of complementary slackness and Karush–Kuhn–Tucker conditions in convex optimization. When applied to the minimization of a Lagrangian over a cone of multipliers, it yields first-order stationarity conditions without requiring any constraint qualification such as linear independence of gradients.

3.7.2 First-Order Optimality Conditions for Mixed Constraints.

We now return to the mixed convex optimization problem with inequality and equality constraints introduced in (3.26). The preceding results establish the existence of Lagrange multipliers and a saddle-point characterization of optimality under convexity and Slater-type regularity, without invoking differentiability. When the problem data are additionally Fréchet differentiable, these global optimality statements admit a refinement in terms of first-order stationarity conditions.

The purpose of the following corollary is to express primal optimality of a solution x_0 in differential form. Specifically, differentiability allows the minimization of the Lagrangian over the primal variable to be characterized by vanishing of its derivative. This yields a system of first-order conditions that generalizes the classical Karush–Kuhn–Tucker conditions to the present abstract setting and clarifies their origin in convex duality rather than constraint qualification arguments.

Corollary 3.15 (Lagrange Multiplier Theorem with First-Order Optimality)**.** *Let all assumptions of Theorem 3.12 hold. Assume in addition that*

$$f : X \to \mathbb{R}, \qquad G : X \to Z, \qquad H : X \to Y$$

are Fréchet differentiable at a point $x_0 \in \Omega$.

Suppose that x_0 attains the optimal value

$$\mu_0 = \inf\{f(x) : x \in \Omega,\ G(x) \preceq 0,\ H(x) = 0\}.$$

Then there exist multipliers

$$z_0^* \in P^{\oplus}, \qquad \lambda_0 \in Y^*$$

such that the following conditions hold:

1. ***Primal feasibility:***

$$G(x_0) \preceq 0, \qquad H(x_0) = 0. \tag{3.39}$$

2. ***Dual feasibility:***

$$z_0^* \in P^{\oplus}. \tag{3.40}$$

3. ***Complementary slackness:***

$$\langle G(x_0), z_0^* \rangle = 0. \tag{3.41}$$

4. ***First-order stationarity:***

$$Df(x_0) + DG(x_0)^* z_0^* + DH(x_0)^* \lambda_0 = 0 \quad \textit{in } X^*. \tag{3.42}$$

Role of differentiability. Corollary 3.15 refines the primal Lagrange multiplier theorem by expressing optimality through first-order conditions. Differentiability is not required for the existence of Lagrange multipliers or for strong duality; those results follow purely from convexity and Slater-type regularity. The role of differentiability is instead to convert global minimization of the Lagrangian into a stationarity condition in the dual space X^*.

Absence of constraint qualifications. No linear independence or rank condition on the constraint mappings is assumed. In particular, linear independence constraint qualifications such as LICQ are not required. The multipliers arise from convex separation arguments rather than from solvability of systems of derivative equations. Constraint qualifications become relevant only when convexity is absent, or when one seeks uniqueness, stability, or sensitivity properties of the multipliers beyond existence.

Geometric meaning of complementary slackness. The complementary slackness condition

$$\langle G(x_0), z_0^* \rangle = 0$$

has a direct geometric interpretation. It expresses orthogonality between the dual multiplier z_0^* and the active face of the cone P at the point $G(x_0)$. Equivalently, it characterizes the supporting hyperplane that separates the epigraph of the Lagrangian from the feasible region in the product space $X \times Z$.

Equality versus inequality constraints. Equality constraints and inequality constraints play fundamentally different roles in the dual formulation. Equality constraints give rise to unrestricted multipliers in the dual space, reflecting the two-sided nature of equality restrictions. In contrast, inequality constraints induce cone-restricted multipliers and lead to complementary slackness conditions. This distinction is intrinsic and cannot be removed by reformulating equalities as pairs of inequalities without destroying regularity.

Connection to classical KKT conditions. In finite-dimensional settings with smooth data, the conditions in Corollary 3.15 reduce to the classical Karush–Kuhn–Tucker system. From this viewpoint, KKT theory is a differentiable specialization of convex duality rather than its foundation. The saddle-point framework provides the conceptual basis, while differentiability merely supplies a convenient representation.

Interpretation as a saddle point. The first-order stationarity condition

$$Df(x_0) + DG(x_0)^* z_0^* + DH(x_0)^* \lambda_0 = 0$$

characterizes x_0 as the primal component of a saddle point of the Lagrangian. Primal feasibility, dual feasibility, complementary slackness, and stationarity are thus unified as consequences of the saddle-point property, rather than as independent optimality requirements.

4 Case Study: Optimal Control with Terminal Inequality Constraints

Consider a linear time-varying control system governed by

$$\dot{x}(t) = A(t)x(t) + b(t)u(t), \qquad t \in [t_0, t_1], \tag{4.1}$$

where $x(t) \in \mathbb{R}^n$ is the state, $A(t) \in \mathbb{R}^{n\times n}$, $b(t) \in \mathbb{R}^{n\times 1}$ are continuous matrix-valued functions, and $u(t) \in \mathbb{R}$ is a scalar control input. The initial state $x(t_0)$ is given.

The objective is to minimize the quadratic control effort

$$J(u) = \frac{1}{2}\int_{t_0}^{t_1} u^2(t)\, dt, \tag{4.2}$$

subject to the terminal inequality constraint

$$x(t_1) \succeq c, \tag{4.3}$$

where $c \in \mathbb{R}^n$ is a prescribed vector and the inequality is understood componentwise.

Let $\Phi(t, s)$ denote the fundamental solution matrix of the homogeneous system $\dot{x}(t) = A(t)x(t)$. Under continuity assumptions on $A(t)$ and $b(t)$, the solution of (4.1) at time t_1 can be expressed as

$$x(t_1) = \Phi(t_1, t_0)x(t_0) + \int_{t_0}^{t_1} \Phi(t_1, t)b(t)u(t)\, dt. \tag{4.4}$$

Define the linear operator $K : L^2[t_0, t_1] \to \mathbb{R}^n$ by

$$Ku := \int_{t_0}^{t_1} \Phi(t_1, t)b(t)u(t)\, dt, \tag{4.5}$$

and set

$$d := c - \Phi(t_1, t_0)x(t_0). \tag{4.6}$$

Then the terminal constraint (4.3) is equivalent to

$$Ku \succeq d. \tag{4.7}$$

The optimal control problem can now be written as the convex optimization problem

$$\begin{aligned} \min_{u \in L^2[t_0,t_1]} \quad & \frac{1}{2}\int_{t_0}^{t_1} u^2(t)\, dt \\ \text{subject to} \quad & Ku \succeq d. \end{aligned} \tag{4.8}$$

The objective functional is strictly convex and continuous on $L^2[t_0, t_1]$, while the constraint mapping K is linear and takes values in the finite- dimensional space $\mathbb{R}^n$ equipped with the positive orthant $\mathbb{R}^n_+$.

Introduce a Lagrange multiplier $\lambda \in \mathbb{R}^n_+$ associated with the constraint (4.7). The Lagrangian is

$$\mathcal{L}(u, \lambda) = \frac{1}{2}\int_{t_0}^{t_1} u^2(t)\, dt + \lambda^\top (d - Ku). \tag{4.9}$$

The dual function is defined by

$$\psi(\lambda) = \inf_{u \in L^2[t_0,t_1]} \mathcal{L}(u, \lambda), \qquad \lambda \in \mathbb{R}^n_+. \tag{4.10}$$

Expanding the Lagrangian yields

$$\mathcal{L}(u,\lambda) = \int_{t_0}^{t_1} \left(\frac{1}{2}u^2(t) - \lambda^\top \Phi(t_1,t)b(t)u(t) \right) dt + \lambda^\top d.$$

For each t, the integrand is a strictly convex quadratic function of $u(t)$. The minimizing control is therefore given pointwise by

$$u^\star(t) = \lambda^\top \Phi(t_1,t)b(t). \tag{4.11}$$

Substituting (4.11) into (4.9) yields

$$\mu_0 = \max_{\lambda \succeq 0} \left\{ \lambda^\top Q\lambda + \lambda^\top d \right\}, \tag{4.12}$$

where

$$Q = -\frac{1}{2}\int_{t_0}^{t_1} \Phi(t_1,t)b(t)b^\top(t)\Phi^\top(t_1,t)\, dt. \tag{4.13}$$

The matrix Q is negative semidefinite, and the dual problem (4.12) is a concave quadratic maximization problem in $\mathbb{R}^n$.

Let $\lambda^\star$ denote a maximizer of (4.12). Then the corresponding optimal control is given by

$$u^\star(t) = (\lambda^\star)^\top \Phi(t_1,t)b(t). \tag{4.14}$$

The multiplier $\lambda^\star$ admits the interpretation of a sensitivity or shadow price: it coincides with the gradient of the optimal value μ_0 with respect to the terminal target c. Thus, $\lambda^\star$ quantifies the marginal cost of tightening the terminal inequality constraint.

Remark 4.1. This example illustrates how an infinite-dimensional optimal control problem with terminal inequality constraints can be reduced, via convex duality, to a finite-dimensional optimization problem. The derivation relies solely on convexity and duality theory and does not invoke Pontryagin's Maximum Principle or differentiability-based optimality conditions. It provides a clear demonstration of the power of saddle-point methods in infinite-dimensional optimization.

5 Case Study: Optimal Extraction of an Exhaustible Resource

A firm owns an underground oil deposit of known total quantity $\alpha > 0$. The firm chooses an extraction rate

$$x : [0, \infty) \to \mathbb{R}_+,$$

where $x(t)$ represents the rate of oil extraction at time t.

The total amount of oil extracted cannot exceed the available stock:

$$\int_0^\infty x(t)\, dt \leq \alpha. \tag{5.1}$$

The firm earns instantaneous profit $F(x(t))v(t)$ at time t, where:

- $F : [0, \infty) \to \mathbb{R}$ is the profit function,
- $v : [0, \infty) \to \mathbb{R}_+$ is a discount factor.

The firm's objective is to maximize total discounted profit:

$$\max_{x(\cdot)} \quad \int_0^\infty F(x(t))\, v(t)\, dt, \tag{5.2}$$

subject to (5.1) and $x(t) \geq 0$ for all t.

We impose the following standard assumptions.

Profit function. The function F satisfies:

1. $F(0) = 0$,
2. F is strictly increasing and strictly concave on $[0, \infty)$,
3. F is continuously differentiable, with derivative denoted F_x.

Consequently, F_x is continuous and strictly decreasing, and therefore admits a continuous, strictly decreasing inverse $(F_x)^{-1}$ on its range.

Discount factor. The discount factor $v(t)$ is continuous, strictly positive, strictly decreasing, satisfies $v(t) \to 0$ as $t \to \infty$, and is integrable on $[0, \infty)$.

5.1 Finite-horizon restriction

To facilitate the application of differentiability and saddle-point arguments, we first restrict attention to extraction policies x that are continuous and vanish outside a finite interval. Fix

$T > 0$ and consider the problem:

$$\begin{aligned} \max_{x(\cdot)} \quad & \int_0^T F(x(t))\, v(t)\, dt \\ \text{s.t.} \quad & \int_0^T x(t)\, dt \le \alpha, \\ & x(t) \ge 0 \quad \text{for } t \in [0, T], \\ & x \text{ continuous on } [0, T]. \end{aligned} \tag{5.3}$$

The constraint set consists of a single scalar inequality and pointwise nonnegativity constraints.

Introduce a Lagrange multiplier $\lambda \ge 0$ associated with the resource constraint. The Lagrangian is

$$L(x, \lambda) = \int_0^T F(x(t))\, v(t)\, dt - \lambda \left(\int_0^T x(t)\, dt - \alpha \right). \tag{5.4}$$

Since the objective is concave and the constraints are convex, optimal solutions correspond to saddle points of the Lagrangian:

$$\min_{\lambda \ge 0} \max_{x \ge 0} L(x, \lambda).$$

Let (x_0, λ_0) denote an optimal primal–dual pair. Introducing a pointwise multiplier $\mu(t) \ge 0$ for the constraint $x(t) \ge 0$, the Karush–Kuhn–Tucker conditions are:

Primal feasibility

$$x_0(t) \ge 0, \qquad \int_0^T x_0(t)\, dt \le \alpha.$$

Dual feasibility

$$\lambda_0 \ge 0, \qquad \mu(t) \ge 0 \quad \text{for all } t \in [0, T].$$

Complementary slackness

$$\lambda_0 \left(\int_0^T x_0(t)\, dt - \alpha \right) = 0, \qquad \mu(t) x_0(t) = 0 \quad \forall t. \tag{5.5}$$

Stationarity

$$F_x(x_0(t))\, v(t) - \lambda_0 + \mu(t) = 0 \quad \forall t \in [0, T]. \tag{5.6}$$

Because F is increasing and $v(t) > 0$, the resource constraint binds at optimality, so $\int_0^T x_0(t)\, dt =$

α and $\lambda_0 > 0$.

5.2 Characterization of the optimal extraction path

From (5.6) and (5.5), two cases arise.

Active extraction. If $x_0(t) > 0$, then $\mu(t) = 0$ and

$$F_x(x_0(t))\, v(t) = \lambda_0,$$

which implies

$$x_0(t) = (F_x)^{-1}\left(\frac{\lambda_0}{v(t)}\right). \tag{5.7}$$

No extraction. If $x_0(t) = 0$, then

$$\mu(t) = \lambda_0 - F_x(0)v(t) \geq 0,$$

so extraction ceases when $\lambda_0 \geq F_x(0)v(t)$.

Since $v(t)$ is strictly decreasing, there exists a cutoff time $t_0 \in (0, T]$ such that

$$\lambda_0 = F_x(0)v(t_0).$$

Consequently, the optimal extraction policy takes the form

$$x_0(t) = \begin{cases} (F_x)^{-1}\left(\dfrac{F_x(0)v(t_0)}{v(t)}\right), & 0 \leq t \leq t_0, \\ 0, & t_0 \leq t \leq T. \end{cases} \tag{5.8}$$

The cutoff time t_0 is uniquely determined by the binding resource constraint:

$$\int_0^{t_0} (F_x)^{-1}\left(\frac{F_x(0)v(t_0)}{v(t)}\right) dt = \alpha.$$

Define

$$J(t_0) = \int_0^{t_0} (F_x)^{-1}\left(\frac{F_x(0)v(t_0)}{v(t)}\right) dt.$$

The function J is continuous, satisfies $\lim_{t_0 \to 0} J(t_0) = 0$ and $\lim_{t_0 \to \infty} J(t_0) = \infty$, and hence admits a unique solution $J(t_0) = \alpha$.

5.3 Extension to the infinite horizon

Any admissible extraction plan $y(t) \geq 0$ with $\int_0^\infty y(t)\,dt = \alpha$ can be approximated arbitrarily closely (in discounted profit) by a continuous function with compact support. Since $v(t) \to 0$ as $t \to \infty$, truncation beyond a sufficiently large horizon induces an arbitrarily small loss in the objective.

Therefore, the solution (5.8) is optimal for the original infinite-horizon problem.

The optimal policy equalizes discounted marginal profit:

$$F_x(x_0(t))v(t) = \lambda_0 \quad \text{for all } t \text{ with } x_0(t) > 0.$$

Extraction is front-loaded, declining over time, and ceases at a finite endogenous exhaustion time t_0.

The multiplier λ_0 admits a shadow-price interpretation:

$$\lambda_0 = \frac{d}{d\alpha}\left(\sup_{x(\cdot)} \int_0^\infty F(x(t))v(t)\,dt\right),$$

representing the marginal value of an additional unit of oil in the ground.

6 Further Reading

This lecture builds on the foundational treatment in Luenberger [2]. The vector-space perspective developed there provides a unifying framework for optimization problems beyond the classical matrix-based setting. In particular, many problems traditionally formulated in finite dimensions, such as semidefinite programming [3] and conic programming [1], admit natural and often more transparent formulations in abstract vector spaces.

More importantly, the vector-space framework is essential for genuinely infinite-dimensional problems, including optimal control problems and optimization problems governed by partial differential equations. From this viewpoint, finite-dimensional optimization appears as a special case of a more general theory rather than its foundation.

Within this abstract setting, the central results of optimization theory, such as saddle-point theorems, Lagrangian duality, and constraint qualification conditions, extend naturally and retain their conceptual meaning. For additional background on vector spaces, dual spaces, and their role in optimization, the reader is referred to [2].

7 Exercises

Exercise 1 (Optimization in Function Spaces)**.** Find a continuous function $x(\cdot)$ on the interval $[-1, 1]$ that maximizes

$$\int_{-1}^{1} t^3 x(t)\, dt$$

subject to the constraint

$$\int_{-1}^{1} |x(t)|^3\, dt \leq 2.$$

Hint: Use Hölder's inequality on a suitably chosen normed linear space.

Exercise 2 (Dual Optimization Problem)**.** Consider the same setting as in the previous exercise, but now minimize

$$\int_{-1}^{1} t^3 x(t)\, dt$$

subject to

$$\int_{-1}^{1} |x(t)|^3\, dt \leq 2.$$

What is the optimal minimizing solution?

Exercise 3 (ℓ^p Spaces)**.** Suppose $x = \{\xi_1, \xi_2, \ldots\} \in \ell_{p_0}$ for some $1 \leq p_0 \leq \infty$.

(a) Show that $x \in \ell_p$ for all $p \geq p_0$.

(b) Show that

$$\|x\|_\infty := \sup_i |\xi_i| = \lim_{p \to \infty} \|x\|_p.$$

For terminology and notation, see Section 2.10 of [2].

Exercise 4 (Bounded Linear Functionals on L^2)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $L_2(\Omega, \mathbb{P}; [0, 1])$ denote the Hilbert space of second-order random vectors (of dimension n) defined on $[0, 1] \times \Omega$. Consider the mappings

$$g : L_2(\Omega, \mathbb{P}; [0, 1]) \to L_2(\Omega, \mathbb{P}), \qquad f : L_2(\Omega, \mathbb{P}) \to \mathbb{R},$$

defined by

$$g(x) = \int_0^1 b(t) x(t, \omega)\, dt, \qquad b \in L_2[0, 1],$$

and

$$f(y) = \mathbb{E}[y].$$

Let $h = f \circ g : L_2(\Omega, \mathbb{P}; [0,1]) \to \mathbb{R}$. Show that both f and h are bounded linear functionals. *Hint:* See Chapter 5.2 of [2].

Exercise 5 (Dual Space of $C[0,1]$)**.** Let $X = C[0,1]$ equipped with the supremum norm, and let $f \in X^*$ be defined by

$$f(x) = x\left(\tfrac{1}{4}\right) + 3x\left(\tfrac{1}{2}\right) + 2\int_0^1 t\,x(t)\,dt.$$

Compute the operator norm $\|f\|$.

Exercise 6 (Interior points of the nonnegative cone in function spaces)**.** Let $[a,b] \subset \mathbb{R}$ be a compact interval and let

$$P := \{f \mid f \geq 0\},$$

where nonnegativity is interpreted according to the ambient function space.

1. Let $1 \leq p < \infty$ and consider the space $L_p[a,b]$ endowed with the L_p norm. Define

$$P := \{f \in L_p[a,b] \mid f(t) \geq 0 \text{ a.e. on } [a,b]\}.$$

   Prove that P has *empty interior* in $L_p[a,b]$.

2. Consider either $C[a,b]$ endowed with the uniform norm $\|\cdot\|_\infty$ or $L_\infty[a,b]$ endowed with the essential supremum norm. Define

$$P := \{f \mid f(t) \geq 0 \text{ for all } t \in [a,b]\}.$$

   Prove that P has *nonempty interior* in these spaces.

Exercise 7 (Minimum-norm duality via first-order optimality)**.** Let X be a real normed linear space with dual X^*, and let $M \subset X$ be a closed linear subspace. Fix $x \in X$ and consider the primal optimization problem

$$\inf_{m \in M} \|x - m\|. \tag{7.1}$$

1. Show that problem (7.1) is a convex optimization problem and that the feasible set M may be represented as the solution set of a linear equality constraint

$$H(m) = 0,$$

   for a suitable continuous linear mapping $H : X \to Y$.

2. Assume that the infimum in (7.1) is attained at some $m_0 \in M$ and that $x - m_0 \neq 0$. Show that the objective function

$$f(m) := \|x - m\|$$

is Fréchet differentiable at m_0 and compute its derivative.

3. Apply Corollary 3.15 to the problem (7.1) and show that there exists a Lagrange multiplier $x_0^* \in X^*$ such that

$$Df(m_0) + DH(m_0)^* \lambda_0 = 0.$$

4. Deduce that $x_0^* \in M^\perp$ and that $\|x_0^*\| = 1$, where

$$M^\perp := \{x^* \in X^* \mid \langle m, x^* \rangle = 0 \text{ for all } m \in M\}.$$

5. Show that

$$\|x - m_0\| = \langle x, x_0^* \rangle,$$

and conclude that

$$\inf_{m \in M} \|x - m\| = \max_{\substack{x^* \in M^\perp \\ \|x^*\| \leq 1}} \langle x, x^* \rangle.$$

Remark. This exercise shows that the minimum-norm duality theorem arises as a direct consequence of first-order optimality for a convex equality-constrained problem, with the dual functional appearing naturally as a Lagrange multiplier rather than as an independently postulated dual variable.

References

[1] M. F. Anjos and J. B. Lasserre. *Handbook on semidefinite, conic and polynomial optimization*, volume 166. Springer Science & Business Media, 2011. (Cited on p. 38)

[2] D. G. Luenberger. *Optimization by vector space methods*. John Wiley & Sons, 1997. (Cited on pp. 38, 39, 40)

[3] H. Wolkowicz, R. Saigal, and L. Vandenberghe. *Handbook of semidefinite programming: theory, algorithms, and applications*, volume 27. Springer Science & Business Media, 2012. (Cited on p. 38)